

Problem 6.10

Bruce and Robbie each open up new bank accounts at time 0. Bruce deposits 100 into his bank account, and Robbie deposits 50 into his. Each account earns the same annual effective interest rate i .

The amount of interest earned in Bruce's account during the 11th year is equal to X . The amount of interest earned in Robbie's account during the 17th year is also equal to X . Calculate X .

For a principal P , the interest earned during the n th year is

$$P i(1+i)^{n-1}.$$

Therefore, Bruce's interest during the 11th year is

$$X = 100i(1+i)^{10}.$$

Robbie's interest during the 17th year is

$$X = 50i(1+i)^{16}.$$

Since these are equal,

$$100i(1+i)^{10} = 50i(1+i)^{16}.$$

Assuming $i > 0$, divide both sides by $50i(1+i)^{10}$:

$$2 = (1+i)^6.$$

Thus,

$$1+i = 2^{1/6},$$

so

$$i = 2^{1/6} - 1.$$

Now substitute into

$$X = 100i(1+i)^{10}.$$

This gives

$$X = 100(2^{1/6} - 1)(2^{1/6})^{10}.$$

Therefore,

$$\boxed{X = 100(2^{1/6} - 1)2^{5/3} \approx 38.79}.$$